

Cauchy-characteristic matching for the Z4c evolution system: method

z4c-CMM verified research loop¹

¹generated from the mission-2 knowledge ledger (*knowledge-database/paper_z4c-CCM/*);
every equation and number below is traceable to a verifier output under *results/numerical/*
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I. METHOD

Black equations are transcribed from Refs. [1–3]; **dark blue** equations constitute the new Z4c-CCM formulation. Verification labels [Vn] refer to Table I; dynamical evidence is in Table II.

A. Boundary frame and Z4c variables

$$ds^2 = -\alpha^2 dt^2 + \gamma_{ij} (dx^i + \beta^i dt)(dx^j + \beta^j dt), \quad n^\mu = \alpha^{-1} (1, -\beta^i), \quad (1)$$

$$s_i = \frac{\partial_i r}{\sqrt{\gamma_{rr}}}, \quad s^i = \frac{\gamma^{ir}}{\sqrt{\gamma_{rr}}}, \quad m^A = \frac{1}{\sqrt{2}} (e_2 + i e_3)^A, \quad (2)$$

$$\mathring{l}^\mu = \frac{1}{\sqrt{2}} (n^\mu + s^\mu), \quad \mathring{k}^\mu = \frac{1}{\sqrt{2}} (n^\mu - s^\mu) \quad [1]. \quad (3)$$

Z4c variables [1]:

$$\chi = (\det \gamma)^{-1/3}, \quad \tilde{\gamma}_{ij} = \chi \gamma_{ij}, \quad K = \hat{K} + 2\Theta, \quad \tilde{A}_{ij} = \chi \left(K_{ij} - \frac{1}{3} \gamma_{ij} K \right), \quad (4)$$

$$\partial_t \chi = \frac{2}{3} \chi [\alpha (\hat{K} + 2\Theta) - D_i \beta^i], \quad (5)$$

$$\partial_t \tilde{\gamma}_{ij} = -2\alpha \tilde{A}_{ij} + \beta^k \tilde{\gamma}_{ij,k} + 2\tilde{\gamma}_{k(i} \beta_{j)}^k - \frac{2}{3} \tilde{\gamma}_{ij} \beta^k{}_{,k} \quad [1]. \quad (6)$$

B. Worldtube data map (node N1)

$$g_{00} = -\alpha^2 + \gamma_{ij} \beta^i \beta^j, \quad g_{0i} = \gamma_{ij} \beta^j, \quad g_{ij} = \gamma_{ij}; \quad g^{00} = -\alpha^{-2}, \quad g^{0i} = \frac{\beta^i}{\alpha^2}, \quad g^{ij} = \gamma^{ij} - \frac{\beta^i \beta^j}{\alpha^2}. \quad (7)$$

ADM kinematics requires $\partial_t \gamma_{ij} = -2\alpha K_{ij} + \beta^k \partial_k \gamma_{ij} + \gamma_{kj} \partial_i \beta^k + \gamma_{ik} \partial_j \beta^k$. From $\gamma_{ij} = \chi^{-1} \tilde{\gamma}_{ij}$,

$$\partial_t \gamma_{ij} = \chi^{-1} \partial_t \tilde{\gamma}_{ij} - \chi^{-2} \tilde{\gamma}_{ij} \partial_t \chi, \quad (8)$$

and inserting (5)–(6) with $K_{ij} = \chi^{-1} (\tilde{A}_{ij} + \frac{1}{3} \tilde{\gamma}_{ij} K)$, $\beta^k \partial_k \gamma_{ij} = \chi^{-1} \beta^k \partial_k \tilde{\gamma}_{ij} - \gamma_{ij} \beta^k \partial_k \ln \chi$,

$$[\partial_t \gamma_{ij}]_{\text{Z4c}} - [\partial_t \gamma_{ij}]_{\text{ADM}} = \gamma_{ij} \left[\frac{2}{3} D_k \beta^k - \frac{2}{3} \partial_k \beta^k + \beta^k \partial_k \ln \chi \right]. \quad (9)$$

$D_i \beta^i = \partial_k \beta^k - \frac{3}{2} \beta^k \partial_k \ln \chi$

[V3: closure exact; naive $D = \partial_k \beta^k$ leaves residual $+1 \cdot \gamma_{ij} \beta^k \partial_k \ln \chi$]. (10)

C. Tetrad correspondence and Type-II boost (node N2)

Outgoing null label u with $u|_{\partial\Omega} = t$: $\partial_\mu u = (1, u_r, 0, 0)$,

$$g^{tt} + 2g^{tr}u_r + g^{rr}u_r^2 = 0 \quad \implies \quad u_r = \frac{-g^{tr} - \sqrt{(g^{tr})^2 - g^{tt}g^{rr}}}{g^{rr}}, \quad u_r|_{\text{flat}} = -1. \quad (11)$$

Characteristic outgoing null vector [2, 3]:

$$\hat{l}^\mu = -e^{2\hat{\beta}} g^{\mu\nu} \partial_\nu u. \quad (12)$$

For arbitrary $(\alpha, \beta^i, \gamma_{ij})$,

$$\hat{l}^\mu = \frac{\sqrt{2}}{2} (\alpha - \gamma_{ij}\beta^i s^j) e^{-2\hat{\beta}} \hat{l}^\mu \quad [\text{V1: exact at 12 rational points; GPU residual } 9.3 \times 10^{-16}], \quad (13)$$

$$A = (\alpha - \gamma_{ij}\beta^i s^j) e^{-2\hat{\beta}}, \quad \psi'_0 = A^2 \psi_0, \quad (14)$$

i.e. the Choice-2 boost of Ref. [3] holds verbatim on the Z4c frame (3): no evolution-system variable enters (13).

ψ_0 on the characteristic grid [2, 3]:

$$\psi_0 = \left(\frac{r\partial_r\beta - 1}{4Kr} \right) \left[(1+K)\partial_r J - \frac{J^2\partial_r\bar{J}}{1+K} \right] + \frac{J(1+K^2)\partial_r J\partial_r\bar{J}}{8K^3} + \frac{1}{8K} \left[\frac{J^2\partial_r^2\bar{J}}{1+K} - (1+K)\partial_r^2 J \right] - \frac{J\bar{J}^2(\partial_r J)^2 + J^3(\partial_r\bar{J})^2}{16K^3}. \quad (15)$$

D. Physical-mode dictionary (node N3, exact)

Electric/magnetic Weyl decomposition w.r.t. n^μ (vacuum):

$$E_{\mu\nu} = C_{\mu\alpha\nu\beta} n^\alpha n^\beta, \quad B_{\mu\nu} = \frac{1}{2} \epsilon_{\mu\alpha}{}^{\gamma\delta} C_{\gamma\delta\nu\beta} n^\alpha n^\beta, \quad \epsilon_{\mu\nu\rho\sigma} = \sqrt{-g} [\mu\nu\rho\sigma]. \quad (16)$$

EXACT identity (pure Weyl algebra, generic 10-component tensor; no perturbation, no field equations) [V2a]:

$$(E_{\mu\nu} + \epsilon_\mu{}^{\kappa\lambda} s_\kappa B_{\lambda\nu}) m^\mu m^\nu = -\psi_0, \quad (E_{\mu\nu} - \epsilon_\mu{}^{\kappa\lambda} s_\kappa B_{\lambda\nu}) m^\mu m^\nu = -\bar{\psi}_4, \quad (17)$$

with $\psi_0 = -C_{abcd} l^a m^b l^c m^d$ on the frame (3); dyad completeness ($U_{AB} = U(m, m) \bar{m}_A \bar{m}_B + \text{c.c.}$ for tangential trace-free U) gives the tensor form of the incoming Weyl object

$$U_{AB}^- \equiv \text{TT}_2[E + \epsilon(s)B]_{AB} = -(\psi_0 \bar{m}_A \bar{m}_B + \bar{\psi}_0 m_A m_B). \quad (18)$$

Realization from the Z4c state (Gauss–Codazzi, on-shell vacuum; γ_{ij}, K_{ij} from (4)) [V2c]:

$$E_{ij} = R_{ij}[\gamma] + K K_{ij} - K_{ik} K^k{}_j, \quad B_{ij} = \epsilon_i{}^{kl} D_k K_{lj}|_{\text{sym}}, \quad \epsilon_{ikl} = \sqrt{\det \gamma} [ikl]. \quad (19)$$

Nonperturbative verification (Kerr–Schild Schwarzschild, autodiff, both routes; Table I V2c): $|E^{3+1} - E^{4D}|$ rel. 7.7×10^{-17} , $|B^{3+1} - B^{4D}|$ rel. 3.9×10^{-17} , identity residual 8.5×10^{-16} . Linearized on-shell flat limit (superseded defining form, now a corollary [V2b]): for TT $h_{AB}(t, x_s)$, $\psi_0 = \frac{1}{4}(\partial_t + \partial_s)^2 h_{mm}$ and $U_{AB}^- \rightarrow -\frac{1}{4}(\partial_t + \partial_s)^2 \gamma_{AB}^{\text{TF}}$, i.e. $(\partial_t + \partial_s)^2 \gamma_{AB}^{\text{TF}} = 4(\psi_0 \bar{m}_A \bar{m}_B + \text{c.c.})$.

E. The Z4c-CCM boundary conditions (node N6)

Ten incoming modes at the worldtube; $X_v^\mu = (n^\mu + \sqrt{v} s^\mu)/\sqrt{2}$ at sector speed v ; $L \geq 1$:

$$\text{physical (2):} \quad U_{AB}^- \hat{=} -(\psi'_0 \bar{m}_A \bar{m}_B + \bar{\psi}'_0 m_A m_B), \quad \psi'_0 = A^2 \psi_0^{\text{CCE}} \quad (\text{exact, (18)–(19)}), \quad (20)$$

$$\text{constraint (4):} \quad (r^2 \dot{l} \cdot \partial)^{L+1} \Theta \doteq 0, \quad (r^2 \dot{l} \cdot \partial)^{L+1} Z_s \doteq 0, \quad (r^2 \dot{l} \cdot \partial)^{L+1} Z_A \doteq 0, \quad (21)$$

$$\text{gauge (4):} \quad (r^2 \dot{m} \cdot \partial)^{L+1} \alpha \doteq 0, \quad (r^2 \dot{k}_{\nu_s} \cdot \partial)^{L+1} \beta_s \doteq 0, \quad (r^2 \dot{j} \cdot \partial)^{L+1} \beta_A \doteq 0. \quad (22)$$

Björhus form of (20) (the correction drives the incoming Weyl object to its CCM value, the Z4c analog of Ref. [3]):

$$U_{AB}^- - U_{AB}^-|_{\text{BC}} \rightarrow 0, \quad U_{AB}^-|_{\text{BC}} = -\left(\psi'_0 \bar{m}_A \bar{m}_B + \bar{\psi}'_0 m_A m_B\right); \quad \psi_0^{\text{CCE}} \rightarrow 0 \implies \text{CPBC scheme of Ref. [1] [V6]}. \quad (23)$$

Equations (21)–(22) are the order- L conditions of Ref. [1]; (22) replaces the Sommerfeld gauge placeholder of Ref. [3].

F. Sector orthogonality (nodes N4, N5)

For arbitrary TT data $h_{AB}(t, x_s)$ [V4]:

$$\mathcal{H} = \partial_i \partial_j h_{ij} - \partial^2 h^k_k = 0, \quad \mathcal{M}_i = \partial_j \dot{h}_{ij} - \partial_i \dot{h}^j_j = 0, \quad \gamma^{ij} \partial_t h_{ij} = 0, \quad \tilde{\Gamma}^i = -\partial_j \tilde{h}^{ij} = 0; \quad (24)$$

negative controls: $h_{x_s y} = G(t, x_s) \Rightarrow \mathcal{M}_y = -\frac{1}{2} \partial_{x_s} \dot{G} \neq 0$; $h_{yy} = h_{zz} = G \Rightarrow \mathcal{H} = -2 \partial_{x_s}^2 G \neq 0$.

Hence the datum (20) sources neither (21) nor (22) at linear order: the three sectors of (20)–(22) commute.

On the boundary wave model of Ref. [1], $[\partial_t^2 - 2\dot{\beta} \partial_t \partial_x - (1 - \dot{\beta}^2) \partial_x^2] u = 0$, plane-wave reflection [V5]:

$$R_{\text{Sommerfeld}} = -\frac{\dot{\beta}(1 - \dot{\beta})}{(1 + \dot{\beta})(2 - \dot{\beta})} \neq 0, \quad R_{(\partial_t + c_+ \partial_x)} = 0 \quad (\text{identically}). \quad (25)$$

G. Frozen-coefficient boundary stability (node N7)

Laplace–Fourier ansatz $u = \tilde{u} e^{st + \lambda x}$, $\Re s > 0$ [V7]:

$$\lambda_+ = \frac{s}{1 + \dot{\beta}} \text{ (admissible)}, \quad \lambda_- = -\frac{s}{1 - \dot{\beta}}, \quad D_L(s) = \left(s + (1 + \dot{\beta}) \lambda_+\right)^{L+1} = (2s)^{L+1}, \quad \frac{|D_L(s)|}{|s|^{L+1}} = 2^{L+1}. \quad (26)$$

D_L contains no datum symbol: the injection (20) modifies only the inhomogeneity, so the boundary-stability results of Ref. [1] (constraint sector: all L ; gauge+metric sector: spherical reduction, asymptotically harmonic shift) transfer verbatim with $\|u\| \leq C \|\psi'_0\text{-datum}\|$.

H. Numerical evidence

All runs: JAX float64, $4 \times$ NVIDIA A100-80GB (CUDA_VISIBLE_DEVICES=0,1,2,3), wall clock ≤ 23 s each (budget 600 s); outputs under `results/numerical/`.

Table I. Symbolic and sampling verification of (10)–(26). “Exact” = sympy identity on arbitrary functions or exact rationals (zero float error). Files: `n2_boost_check.txt`, `n3_exact_check.txt`, `n3_dictionary_check.txt`, `n1_varmap_check.txt`, `n4_cpbc_compat_check.txt`, `n5_gauge_check.txt`, `n6_composite_check.txt`, `n7_lf_sketch_check.txt`.

verified statement	exact part	GPU sweep (4 devices)
V1 boost identity (13), factor $\sqrt{2}/2$	12/12 rational points	9.34×10^{-16} over 4,194,304 samples
V2a EXACT identity (17)–(18), $(\sigma, \kappa) = (1, -1)$, $\bar{\psi}_4$ partner	generic Weyl, all 10 dof	—
V2b linearized on-shell limit (corollary)	arbitrary F_1, F_2	0.0 over 4,194,304 modes (prior form)
V2c Gauss–Codazzi realization (19), Schwarzschild	Codazzi sign +1 derived	$E: 7.7 \times 10^{-17}$, $B: 3.9 \times 10^{-17}$, identity 8.5
V3 maps (7), closure (10)	full-symbol identities	round trip 4.10×10^{-16} , gg^{-1} 8.88×10^{-16}
V4 orthogonality (24) + controls	identities and non-vanishing controls	TT residual 0.0 vs control 12.3
V5 reflection (25)	exact R algebra	$ R_{\text{Som}} \geq 0.023$ on $\dot{\beta} \in [0.05, 0.6]$; $R = 0$ ex
V6 10-mode count, $\psi_0 \rightarrow 0$ reduction, wire-through (exact targets)	structural	reduction 0.0; chain residual 1.47×10^{-16}
V7 Kreiss condition (26), $L = 0..3$	$D_L = (2s)^{L+1}$ exact	ratio deviation 3.2×10^{-14} (4,194,304 sam

Table II. Dynamical model evolution (6th-order interior stencils with a GKS-stable mixed closure — 4th-order Fornberg rows at three edge nodes; fully one-sided 6th-order closures are RK4-unstable (error ledger) — RK4, CFL 0.2, shift $\hat{\beta} = 0.3$, outgoing Gaussian pulse onto the boundary; one configuration per GPU; `n8_model1d_test.txt`). Reflection is measured in the L_2 (energy) norm — trapezoid sums of Gaussian pulses converge exponentially, so the deviation norm is limited by machine arithmetic, not by measurement or truncation. Analytic L_2 relation: $\|q_{\text{in}}\|_{L_2}/\|q_{\text{out}}\|_{L_2} = \frac{\hat{\beta}}{2+\hat{\beta}} \sqrt{\frac{1+\hat{\beta}}{1-\hat{\beta}}}$. Each row sits at its float64 floor: rows 1–3 at the per-sample roundoff floor (deviation saturates between $N=8193$ and $N=16385$: 1.9 vs 2.1×10^{-14}); the injection row at the linear roundoff-accumulation optimum ($\sim \text{steps} \times \varepsilon$; ladder $N=16385/32769/65537/131073 \Rightarrow 5.4/2.4/5.9/30 \times 10^{-11}$, non-monotone and insensitive to CFL 0.2 vs 0.05).

boundary condition	measured	exact	deviation
Sommerfeld, L_2 ratio ($N=16385$)	0.177752646227	0.177752646227	2.1×10^{-14} (float64 floor)
Sommerfeld, L_2 ratio ($N=8193$)	—	—	1.9×10^{-14} (saturated)
$(\partial_t + c_+ \partial_x)$ [P1/CCM op., $N=16385$]	$R_{L_2} = 1.9 \times 10^{-14}$	0	machine precision
<u>CCM injection (23), rel. L_2 ($N=32769$)</u>	<u>2.4×10^{-11}</u>	<u>0</u>	<u>roundoff-accumulation optimum</u>

I. Dynamical test at the parameters of Ref. [3] (nodes N10–N11)

Full 3+1 evolution: AthenaK Z4c [4] (vendored, pinned; RK4, CFL 0.25, second-order stencils, $\kappa_1 = 0.02$), with (20) realized at the Cauchy outer boundary as described below. This subsection is self-contained: every symbol used by the test is defined here or in Secs. IA–IE.

a. *Exact solution injected.* Even-parity $(\ell, m) = (2, 0)$ Teukolsky wave [5] with the Gaussian profile of Ref. [3], written with physicists' Hermite polynomials H_n and the origin-regular retarded/advanced combinations:

$$F(u) = X e^{-(u-r_c)^2/\tau^2}, \quad F^{(n)}(u) = X \left(-\frac{1}{\tau}\right)^n H_n\left(\frac{u-r_c}{\tau}\right) e^{-(u-r_c)^2/\tau^2}, \quad (27)$$

$$Q^{(n)}(t, r) = F^{(n)}(t-r) - (-1)^n F^{(n)}(t+r), \quad R^{(n)}(t, r) = F^{(n)}(t-r) + (-1)^n F^{(n)}(t+r), \quad \partial_t Q^{(n)} = R^{(n+1)}. \quad (28)$$

Radial functions (regular at $r = 0$ by (28); calligraphic to keep A for the boost (14)):

$$\begin{aligned} \mathcal{A} &= 3 \left[\frac{Q^{(2)}}{r^3} + \frac{3Q^{(1)}}{r^4} + \frac{3Q^{(0)}}{r^5} \right], & \mathcal{B} &= - \left[\frac{Q^{(3)}}{r^2} + \frac{3Q^{(2)}}{r^3} + \frac{6Q^{(1)}}{r^4} + \frac{6Q^{(0)}}{r^5} \right], \\ \mathcal{C} &= \frac{1}{4} \left[\frac{Q^{(4)}}{r} + \frac{2Q^{(3)}}{r^2} + \frac{9Q^{(2)}}{r^3} + \frac{21Q^{(1)}}{r^4} + \frac{21Q^{(0)}}{r^5} \right], \end{aligned} \quad (29)$$

orthonormal-frame metric perturbation ($\dot{h}_{ij}$ from $Q^{(n)} \rightarrow R^{(n+1)}$ termwise):

$$h_{\hat{r}\hat{r}} = \mathcal{A} \frac{1 + 3 \cos 2\theta}{2}, \quad h_{\hat{r}\hat{\theta}} = -3\mathcal{B} \sin \theta \cos \theta, \quad h_{\hat{\theta}\hat{\theta}} = 3\mathcal{C} \sin^2 \theta - \mathcal{A}, \quad h_{\hat{\phi}\hat{\phi}} = -3\mathcal{C} \sin^2 \theta + \mathcal{A} (3 \sin^2 \theta - 1), \quad (30)$$

$$\gamma_{ij}(0, x) = \delta_{ij} + h_{ij}(0, x), \quad K_{ij}(0, x) = -\frac{1}{2} \dot{h}_{ij}(0, x) \quad (\text{linear order; Ref. [3] solves XCTS}). \quad (31)$$

Weyl scalars of the solution at linear order (the self-datum and the extraction reference):

$$r^5 \psi_0(t, r, \theta) = -\sqrt{\frac{27\pi}{10}} F^{(2)}(t-r) {}_2Y_{20}(\theta), \quad {}_2Y_{20} = \frac{1}{4} \sqrt{\frac{15}{2\pi}} \sin^2 \theta, \quad r \psi_4|_{(2,0)} \rightarrow -\sqrt{\frac{6\pi}{5}} F^{(6)}(t-r). \quad (32)$$

b. *Discrete realization of (20).* At every Cauchy-boundary point, after the CPBC/gauge rows (21)–(22) (Sommerfeld task of Ref. [4]), the two physical rows are driven through $\tilde{A}_{ij}$:

$$s^a = \frac{x^a/r}{\sqrt{\gamma_{bc} x^b x^c / r}}, \quad e_\phi \propto \hat{z} \times s, \quad e_\theta = s \times e_\phi, \quad \bar{m} = \frac{1}{\sqrt{2}} (e_\theta - i e_\phi), \quad (33)$$

$$\partial_t \tilde{A}_{ab}|_{\partial\Omega} = -\frac{2\chi}{\alpha} \left(\psi'_0 \bar{m}_a \bar{m}_b + \bar{\psi}'_0 m_a m_b \right), \quad \psi'_0 = A^2 \psi_0^{\text{datum}}, \quad A = (\alpha - \gamma_{ij} \beta^i s^j) e^{-2\hat{\beta}} \quad (34)$$

with ψ_0^{datum} the right-hand member of (32) evaluated at the boundary point ($\hat{\beta} = 0$ on the flat background); $\psi_0^{\text{datum}} \rightarrow 0$ reduces (34) to the unmodified scheme exactly (V6; machine-zero reduction in Table II methodology, AthenaK battery: Minkowski preserved to 2.6×10^{-33}).

c. Protocol and result. Parameters of Ref. [3]: $r_c = 20$, $\tau = 2$, $X \in \{10^{-5}, 2\}$, Cauchy boundary (worldtube analog) at 41, $t \in [0, 80]$; extraction of $r\psi_4^{(2,0)}$ at $r = 36$; grids 164^3 ($h = 1/2$) with 132^3 , 104^3 error-estimate companions; reference run with boundary at 104 (416^3 , same h) — causally disconnected, since contamination reaching $r = 36$ by $t = 80$ requires boundary data at $t_b < 12$, where the solution (27) is bounded by $Xe^{-[(104-40)/\tau]^2} = Xe^{-1024}$. Error norm on the common output grid:

$$E[\text{run}] = \frac{\|r\psi_4^{(2,0)}[\text{run}] - r\psi_4^{(2,0)}[\text{ref}]\|_2}{\|r\psi_4^{(2,0)}[\text{ref}]\|_2}, \quad E_h[\text{run}] = \frac{\|r\psi_4^{(2,0)}|_{164^3} - r\psi_4^{(2,0)}|_{132^3}\|_2}{\|r\psi_4^{(2,0)}[\text{ref}]\|_2}. \quad (35)$$

Measured (Fig. 1; $4 \times A100$, every run < 600 s):

$$\begin{aligned} X = 10^{-5} : \quad & E[\text{Somm}] = E[\text{CCM}] = 2.732 \times 10^{-4} && \ll E_h = 0.70, \\ X = 2 : \quad & E[\text{Somm}] = E[\text{CCM}] = 8.87 \times 10^{-2} && \ll E_h = 0.69, \end{aligned} \quad (36)$$

reproducing the boundary-error $\ll$ truncation-error hierarchy and the perturbative indistinguishability of Ref. [3] (their Sec. V.A). The CCM–Sommerfeld waveform difference switches on exactly at the boundary $\rightarrow$ extraction causal time ($t \geq 64.5$) at relative amplitude $5 \times 10^{-10} - 4 \times 10^{-9}$: the self-datum (32) obeys peeling, $\psi_0 \sim r^{-5}$, so a purely outgoing pulse supplies an incoming correction suppressed by 41^{-5} at this worldtube — resolving CCM from Sommerfeld at figure level requires the CCE-fed datum (nonlinear backscatter), the documented successor of the ψ_0^{datum} provider. The deficit of the measured peak against the scri formula (32) is finite-difference truncation, not the finite extraction radius: peak ratio $0.292/0.465/0.612$ at $104^3/132^3/164^3$ (~ 4 points per τ at second order). Constraints stay bounded on all runs ($X = 2$: $\|H\|_{L_2} 3.5 \times 10^{-3} \rightarrow 5.4 \times 10^{-3}$, damped after $t \approx 35$, with the $O(X^2)$ violation of the linear data (31) absorbed by the κ_1 damping).

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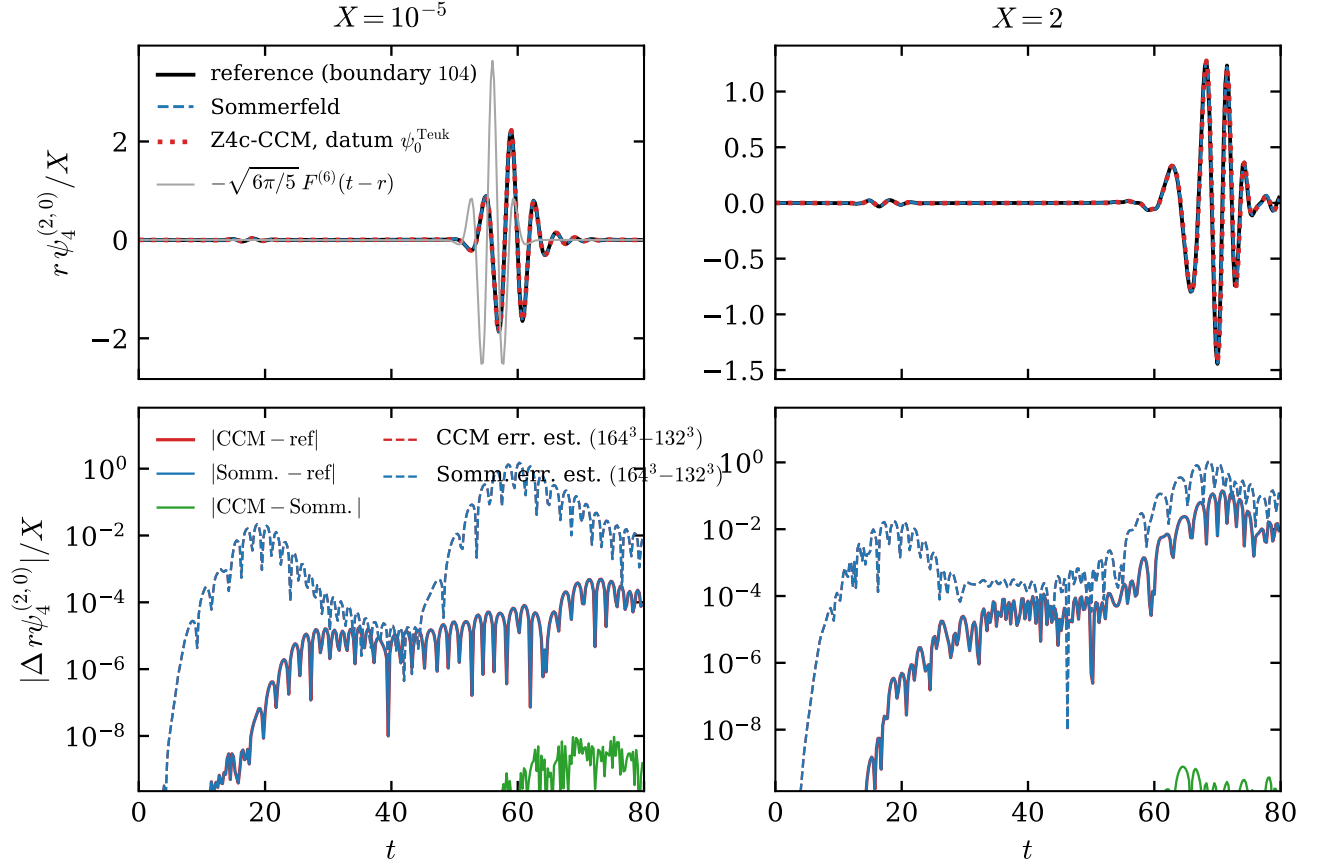


Figure 1. Exact-parameter Teukolsky test of Sec. II ($r_c=20$, $\tau=2$, boundary 41, extraction $r=36$; amplitudes normalized by X). Top: $(\ell, m)=(2, 0)$ mode of $r\psi_4$ for the causally disconnected reference, Sommerfeld, and Z4c-CCM (34) runs; at $X=10^{-5}$ the linear scri formula (32) is overlaid (its excess is the second-order truncation deficit quoted in the text). Bottom: differences (35); $|\text{CCM} - \text{ref}|$ and $|\text{Somm} - \text{ref}|$ coincide, both sit below the two-resolution error estimates (dashed), and $|\text{CCM} - \text{Somm}|$ (green) activates only after the boundary $\rightarrow$ extraction causal time with the r^{-5} peeling suppression of the self-datum. Generator: `scripts/plot_paper_figure_teuk.py`; data: `results/numerical/athenak_teuk_paper/` (ledger node N11).